

Owen M. Hoffman

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EDUCATION

University of Maryland, College Park | College Park, MD

Aug. 2026 – Present

Ph.D. in Computer Science

- GAMMA Lab; research with Prof. Dinesh Manocha.

Swarthmore College | Swarthmore, PA

Aug. 2022 – May 2026

B.A. in Computer Science & B.S. in Engineering

- GPA: 3.90/4.00

RESEARCH INTERESTS

Safe, human-aware embodied AI; socially aware robot navigation; multimodal human behavior understanding; learning-informed planning and control.

PUBLICATIONS

Refereed Conference Papers

- **Hoffman, O.**, Peng, K., Kamal, S., You, Z., & Venkatagiri, S. (2026). *ScamPilot: Simulating Conversations with LLMs to Protect Against Online Scams*. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. DOI

Workshop Papers

- **Hoffman, O.**, Peng, K., You, Z., Kamal, S., & Venkatagiri, S. (2025). *Promoting Online Safety by Simulating Unsafe Conversations with LLMs*. Position paper, *ACM CUI Workshop on Personas Evolved: Designing Ethical LLM-Based Conversational Agent Personalities*. arXiv:2507.22267

RESEARCH EXPERIENCE

Graduate Researcher – GAMMA Lab

Aug. 2026 – Present

University of Maryland, Department of Computer Science – Prof. Dinesh Manocha

College Park, MD

- Investigating safe, human-aware embodied AI, with an emphasis on socially aware robot navigation in environments shared with people.
- Exploring how multimodal cues about human behavior can inform robot perception, planning, and decision-making.

Research Assistant – Collective Resilience Lab

Jan. 2025 – Aug. 2026

Swarthmore College, Computer Science – Prof. Sukrit Venkatagiri

Swarthmore, PA

Privacy Navigator

Sep. 2025 – Aug. 2026

- Developed a pipeline to extract website privacy policies and use an LLM to identify and score disclosures related to data collection and use, with a second LLM pass triggered when the initial web-scraping step fails.
- Built a system that extracts URLs and email contacts from privacy policies and uses an LLM to identify actionable steps users can take to improve their online privacy.
- Designed the interface and evaluation methodology for communicating website privacy practices and recommended actions to users.

ScamPilot: Simulating Conversations with LLMs to Protect Against Online Scams Jan. 2025 – Sep. 2025

- Designed and implemented an LLM-agent system that simulated adaptive scam conversations to investigate whether interactive training improves users' ability to recognize and respond to online scams.
- Developed the structured prompting and experimental pipeline used to maintain controlled agent behavior across experimental conditions.
- Conducted a controlled user study and implemented the statistical analysis pipeline used to evaluate effects on scam recognition, response efficacy, and self-efficacy.
- **Outcome:** First-author paper published at ACM CHI 2026; position paper accepted at the ACM CUI Personas Workshop 2025.

HONORS & AWARDS

- University of Maryland Dean’s Fellowship** — Graduate fellowship supporting doctoral study in Computer Science (2026).
- Swarthmore College Summer Research Fellowship** — Supported full-time undergraduate research in Human–Computer Interaction (Summer 2025).
- Tau Beta Pi**, Engineering Honor Society — Elected 2026.
- Sigma Xi**, Scientific Research Honor Society — Associate Member, elected 2025.
- Sigma Xi Travel Grant** — Supported presentation at the ACM CUI Workshop on Personas Evolved (2025).

TEACHING

- Teaching Assistant – CMSC 216: Introduction to Computer Systems** Aug. 2026 – Present
University of Maryland, Department of Computer Science *College Park, MD*
- Support instruction in systems programming concepts, including C programming, memory management, debugging, and low-level computer systems.
 - Hold office hours and provide technical guidance to students on programming assignments and course material.
- Grader & Lab Teaching Assistant** Sep. 2024 – May 2026
Swarthmore College, Department of Engineering *Swarthmore, PA*
- Supported *ENGR 021: Computer Engineering Fundamentals* and *ENGR 012: Linear Physical Systems Analysis* across multiple semesters through grading, rubric development, and written feedback.
 - Provided hands-on debugging and technical guidance to student teams during computer engineering laboratories.
 - Coordinated with instructors and other teaching staff to refine assessments and course materials.

SELECTED PROJECTS

- Senior Capstone: Trash vs. Recycling Robot** | *ROS 2, Python, CasADi, MPC* | GitHub
- Implemented a CasADi-based model predictive controller for ROS 2 mobile robot navigation, including obstacle-avoidance constraints and optimization of linear and angular velocity over a finite prediction horizon.
 - Collaborated with a three-person team to develop the robot’s broader perception-to-control pipeline for approaching seated users, collecting deposited items, and routing trash and recycling to separate onboard bins.
- Autonomous TurtleBot Cone-Course Navigation** | *ROS, Python*
- Integrated camera-based perception, wheel-encoder odometry, and ROS frame transforms for autonomous cone-course navigation.
 - Implemented a pure-pursuit controller with acceleration filtering and a search behavior for recovering when visual gates were lost.
 - Integrated bumper and cliff sensor feedback to enforce emergency-stop behavior during navigation.

TECHNICAL SKILLS

- Robotics & Perception:** ROS/ROS 2, Nav2, OpenCV, mobile robot navigation, path planning, model predictive control, camera-based perception
- Machine Learning & AI:** PyTorch, TensorFlow/Keras, scikit-learn, neural networks, large language models, multimodal learning
- Programming:** Python, C/C++, MATLAB, R, TypeScript/JavaScript, SQL
- Data & Systems:** NumPy, pandas, Git, Docker, Linux, AWS

INDUSTRY EXPERIENCE

- Software Developer Intern** Jun. 2024 – Aug. 2024
Plurilock Security Inc. *Remote*
- Developed and iterated on a TypeScript/React user interface for a cybersecurity product.
 - Built and integrated more than ten REST API service calls with JWT-based authorization and dynamic client-side data updates.
 - Contributed through code reviews, GitFlow-based development, testing, and QA in an agile engineering team.

ACTIVITIES & INTERESTS

Athletics: NCAA Division III Varsity Men's Lacrosse, Swarthmore College (2022–2026); team ranked as high as No. 4 nationally in the USILA poll.

Interests: Running, specialty coffee, downhill skiing.